Lecture 7 - SVM and Kernels

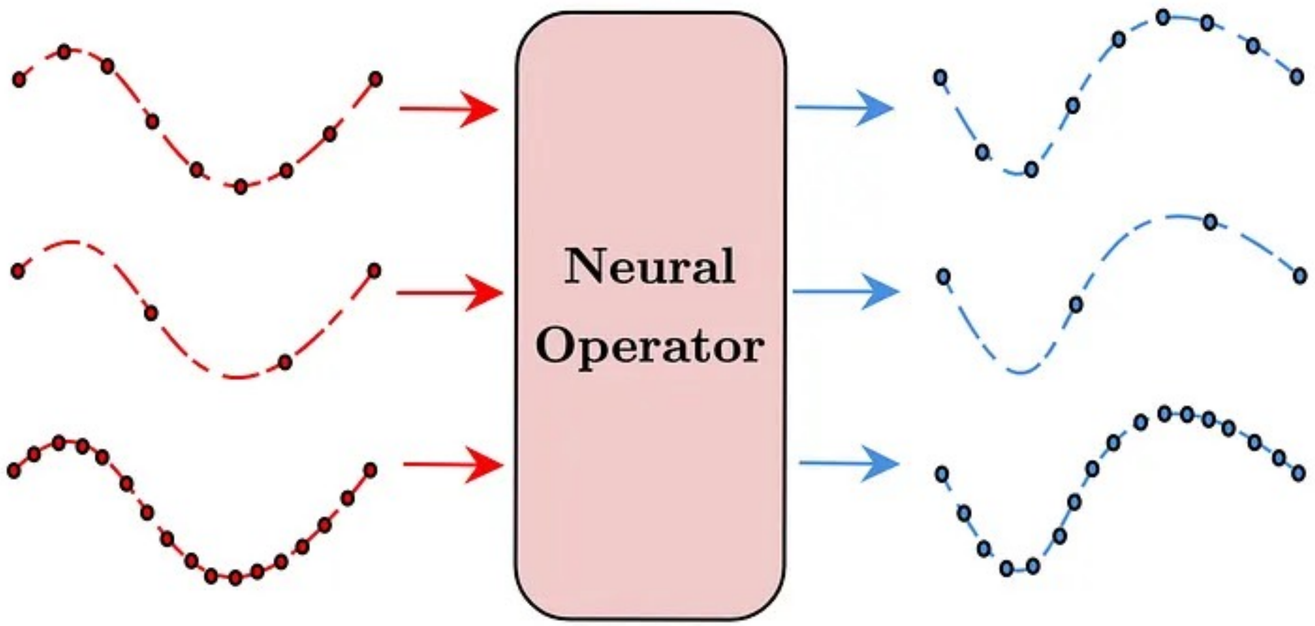
ECE 447: Data Analysis and Machine Learning for Engineers

Bahareh Tolooshams, Ph.D.
Assistant Professor,
University of Alberta
btolooshams@ualberta.ca
https://btolooshams.github.io

February 03, 2026

Mapping between Two Function Spaces

This slide is informational, excluded from assessment.



Fourier Neural Operator for Parametric Partial Differential Equations
Z Li, NB Kovachki, K Azizzadenesheli, B Lui, K Bhattacharya, A Stuart, ...
International Conference on Learning Representations

Neural operator: Learning maps between function spaces with applications to pdes
N Kovachki, Z Li, B Liu, K Azizzadenesheli, K Bhattacharya, A Stuart, ...
Journal of Machine Learning Research 24 (89), 1-97

https://medium.com/@bogdan.raonke/operator-learning-convolutional-neural-operators-for-robust-and-accurate-learning-of-pdes-ebbc43b57434

This Lecture Covers ...

- Revisit SVM
  - KKT conditions
- Kernels
  - Regression in feature space
  - Kernel regression
  - Kernel trick!

This Lecture Covers ...

- Revisit SVM
  - KKT conditions

- Kernels
  - Regression in feature space
  - Kernel regression
  - Kernel trick!

Lagrangian and Optimality Conditions for SVM (Separable Case)

From last lecture.

Lagrangian of the objective is

$$L(\boldsymbol{\theta}_1, \boldsymbol{\theta}_0, \boldsymbol{\lambda}) = \frac{1}{2}\|\boldsymbol{\theta}_1\|_2^2 \; - \; \sum_{i=1}^{n} \lambda_i \left(y_i(\boldsymbol{\theta}_1^\top \boldsymbol{x}_i + \theta_0) - 1 \right), \quad \boldsymbol{\lambda} \in \mathbb{R}^n$$

The optimality conditions are

(I) $\quad \nabla_{\boldsymbol{\theta}_1} L(\boldsymbol{\theta}_1, \boldsymbol{\theta}_0, \boldsymbol{\lambda}) = 0 \quad \Leftrightarrow \quad \hat{\boldsymbol{\theta}}_1 = \sum_{i=1}^{n} \lambda_i y_i \boldsymbol{x}_i$

(II) $\quad \nabla_{\theta_0} L(\boldsymbol{\theta}_1, \boldsymbol{\theta}_0, \boldsymbol{\lambda}) = 0 \quad \Leftrightarrow \quad \sum_{i=1}^{n} \lambda_i y_i = 0$

(III) $\quad \begin{cases} \lambda_i \left(y_i(\boldsymbol{\theta}_1^\top \boldsymbol{x}_i + \theta_0) - 1 \right) = 0 & \text{for } i = 1, \ldots, n \\ \lambda_i \geq 0 \end{cases}$

If $\lambda_i \neq 0$, *then* $y_i(\boldsymbol{\theta}_1^\top \boldsymbol{x}_i + \theta_0) = 1$

Karush–Kuhn–Tucker (KKT) Conditions (for λ_i)

$$L(\boldsymbol{\theta}_1, \boldsymbol{\theta}_0, \boldsymbol{\lambda}) = \frac{1}{2}\|\boldsymbol{\theta}_1\|_2^2 \; - \; \sum_{i=1}^{n} \lambda_i \left(y_i(\boldsymbol{\theta}_1^\top \boldsymbol{x}_i + \theta_0) - 1 \right), \quad \boldsymbol{\lambda} \in \mathbb{R}^n$$

Stationarity Condition (from gradient being zero)

$$y_i(\boldsymbol{\theta}_1^\top \boldsymbol{x}_i + \theta_0) - 1 = 0$$

Complementary Slackness (this is applied to inequality constraints)

$$\lambda_i \left(y_i(\boldsymbol{\theta}_1^\top \boldsymbol{x}_i + \theta_0) - 1 \right) = 0$$

Dual Feasibility

$$\lambda_i \geq 0$$

Closer Look into Complementary Slackness for SVM

1. For **correctly classified** points: $y_i(\boldsymbol{\theta}_1^\top \boldsymbol{x}_i + \theta_0) - 1 > 0$.
Hence, to reach stationarity condition, $\lambda_i = 0$.

To make sure inactive constraints are not impacting the objective

2. For **support vectors**: $y_i(\boldsymbol{\theta}_1^\top \boldsymbol{x}_i + \theta_0) - 1 = 0$. *Hence, the constraint is **active.** Hence, the multiplier can be non-zero $\lambda_i \neq 0$.*

Complementary Slackness (this is applied to inequality constraints)

$$\lambda_i \left(y_i(\boldsymbol{\theta}_1^\top \boldsymbol{x}_i + \theta_0) - 1 \right) = 0$$

The Lagrangian for Non-Separable SVM (non-separable case)

From last lecture.

$$L(\boldsymbol{\theta}_1, \boldsymbol{\theta}_0, \boldsymbol{\zeta}, \boldsymbol{\lambda}, \boldsymbol{\gamma}) = \frac{1}{2}\|\boldsymbol{\theta}_1\|_2^2 + C\sum_{i=1}^{n}\zeta_i - \sum_{i=1}^{n}\gamma_i\zeta_i - \sum_{i=1}^{n}\lambda_i(y_i(\boldsymbol{\theta}_1^\top \boldsymbol{x}_i + \theta_0) - 1 + \zeta_i)$$

The optimality conditions are

(I) $\nabla_{\boldsymbol{\theta}_1} L(\boldsymbol{\theta}_1, \boldsymbol{\theta}_0, \boldsymbol{\zeta}, \boldsymbol{\lambda}, \boldsymbol{\gamma}) = 0 \quad \Leftrightarrow \quad \hat{\boldsymbol{\theta}}_1 = \sum_{i=1}^{n} \lambda_i y_i \boldsymbol{x}_i$

(II) $\nabla_{\theta_0} L(\boldsymbol{\theta}_1, \boldsymbol{\theta}_0, \boldsymbol{\zeta}, \boldsymbol{\lambda}, \boldsymbol{\gamma}) = 0 \quad \Leftrightarrow \quad \sum_{i=1}^{n} \lambda_i y_i = 0$

(III) $\nabla_{\zeta_i} L(\boldsymbol{\theta}_1, \boldsymbol{\theta}_0, \boldsymbol{\zeta}, \boldsymbol{\lambda}, \boldsymbol{\gamma}) = 0 \quad \Leftrightarrow \quad C - \lambda_i - \gamma_i = 0$

(IV) $\begin{cases} \lambda_i \left(y_i(\boldsymbol{\theta}_1^\top \boldsymbol{x}_i + \theta_0) - 1 + \zeta_i\right) = 0 \\ \lambda_i \geq 0 \end{cases}$ for $i = 1, \ldots, n$

(V) $\begin{cases} \gamma_i \zeta_i = 0 \\ \gamma_i \geq 0 \end{cases}$ for $i = 1, \ldots, n$

Non-Separable SVM Remarks

From last lecture.

- For all points on the correct side of the decision boundary:

$$\zeta_i = 0 \quad \text{and} \quad y_i(\boldsymbol{\theta}_1^\top \boldsymbol{x}_i + \theta_0) > 1 \quad \Rightarrow \quad \lambda_i = 0$$

- The support vectors ($\lambda_i \neq 0$) are points on the border, inside the margin, and on the wrong side of the decision boundary.

- For all points inside the margin or the wrong side:

$$\zeta_i > 0 \quad \text{and} \quad \lambda_i = C \quad \Rightarrow \quad \gamma_i = 0$$

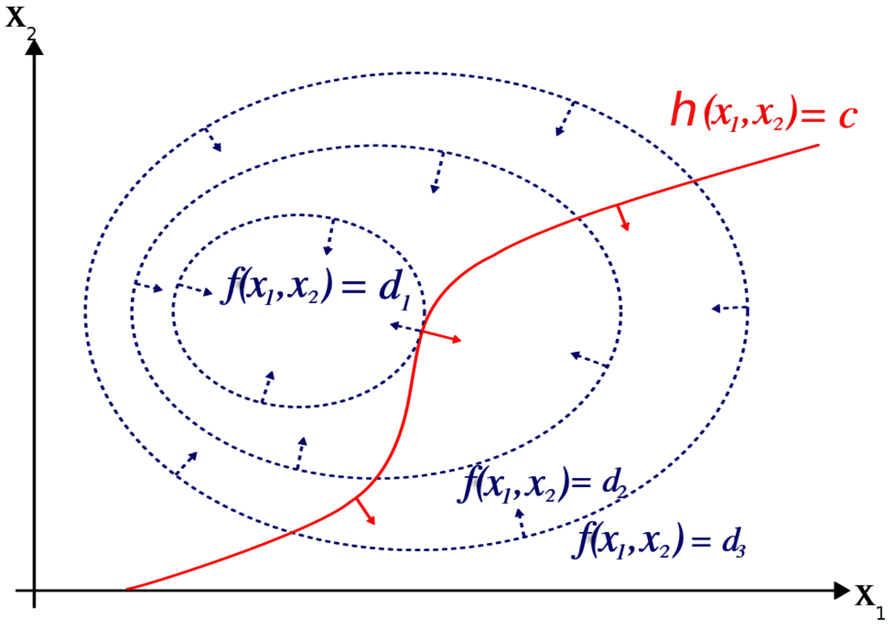
- For support vectors on the margin border:

$$\zeta_i = 0 \quad \text{and} \quad 0 \leq \lambda_i \leq C \quad \Rightarrow \quad \gamma_i \geq 0$$

General Visualization of Optimization with Constraints

This slide is informational, excluded from assessment.

Equality constraint



https://core-robotics.gatech.edu/2023/08/30/bootcamp-summer-2020-week-karush-kuhn-tucker-kkt-conditions/

Inequality constraint

done on the board

Constrained Optimization

https://www.ee.iitm.ac.in/uday/notes/opt/co1.html

This Lecture Covers ...

- Revisit SVM
  - KKT conditions

- Kernels
  - Regression in feature space
  - Kernel regression
  - Kernel trick!

Polynomial Regression Models

In previous lectures, we have studied models like $y_i = \boldsymbol{\theta}_1^\top \boldsymbol{x}_i + \theta_0 \leftarrow$ *linear in* $\boldsymbol{x}_i$.

Given a set of data points $\{x_i, y_i\}_{i=1}^n$, with $y_i \in \mathbb{R}$ and $x_i \in \mathbb{R}$,

$$y_i = \theta_0 + \theta_1(\sqrt{2}x_i) + \theta_2 x_i^2 + \epsilon_i$$

$\epsilon_i \sim \mathcal{N}(0,\sigma^2)$

$i = 1,\ldots,n$

find a feature space $\Phi(x)$ such that the regression is linear in there.

$$y_i = [\theta_0, \theta_1, \theta_2]^\top \begin{bmatrix} 1 \\ \sqrt{2}x_i \\ x_i^2 \end{bmatrix} + \epsilon_i \quad \Rightarrow$$

$\Phi : \mathbb{R} \to \mathbb{R}^3$, where

$$x \to \Phi(x) = \begin{bmatrix} 1 \\ \sqrt{2}x \\ x^2 \end{bmatrix}$$

This is called a **feature map**.

Regularized Least Squares in Feature Space

Reminder from linear regression in **original** space.

$$\hat{\boldsymbol{\theta}} = (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{X}^\top \boldsymbol{y}$$

the **Least Square (LS)** estimate.

Given a set of data points $\{\boldsymbol{x}_i, y_i\}_{i=1}^n$, find the parameters $\boldsymbol{\theta}$ such that for new point $\boldsymbol{x}$, one predict $\hat{y}$ as

$$\hat{y} = \boldsymbol{\theta}^\top \Phi(\boldsymbol{x})$$

$$\hat{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}} \ \frac{1}{2}\|\boldsymbol{y} - \Phi(\boldsymbol{X})\boldsymbol{\theta}\|_2^2 \ + \ \frac{\lambda}{2}\|\boldsymbol{\theta}\|_2^2$$

$$\nabla_{\boldsymbol{\theta}} \frac{1}{2}(\boldsymbol{y} - \Phi(\boldsymbol{X})\boldsymbol{\theta})^\top(\boldsymbol{y} - \Phi(\boldsymbol{X})\boldsymbol{\theta}) + \ \frac{\lambda}{2}\boldsymbol{\theta}^\top\boldsymbol{\theta} = 0$$

$$\Leftrightarrow \quad \Phi(\boldsymbol{X})^\top(\boldsymbol{y} - \Phi(\boldsymbol{X})\boldsymbol{\theta}) + \lambda\boldsymbol{\theta} = 0$$

$$\Leftrightarrow \quad (\Phi(\boldsymbol{X})^\top\Phi(\boldsymbol{X}) + \lambda\boldsymbol{I})\boldsymbol{\theta} = \Phi(\boldsymbol{X})^\top\boldsymbol{y}$$

$$\Rightarrow$$

$$\hat{\boldsymbol{\theta}} = (\Phi(\boldsymbol{X})^\top\Phi(\boldsymbol{X}) + \lambda\boldsymbol{I})^{-1}\Phi(\boldsymbol{X})^\top\boldsymbol{y}$$

the **Regularized Least Square (LS)** estimate in the **feature space**.

Regression in Feature Space (Regularized Least Square Case)

Given a set of data points $\{\boldsymbol{x}_i, y_i\}_{i=1}^n$, the least square estimate is

$$\hat{\boldsymbol{\theta}} = (\Phi(\boldsymbol{X})^\top \Phi(\boldsymbol{X}) + \lambda \boldsymbol{I})^{-1} \Phi(\boldsymbol{X})^\top \boldsymbol{y}$$

where

$$\boldsymbol{X} = \begin{bmatrix} -\boldsymbol{x}_1^\top - \\ -\boldsymbol{x}_2^\top - \\ \vdots \\ -\boldsymbol{x}_n^\top - \end{bmatrix} \quad \text{and} \quad \Phi(\boldsymbol{X}) = \begin{bmatrix} 1 & \sqrt{2}\boldsymbol{x}_1 & \boldsymbol{x}_1^2 \\ 1 & \sqrt{2}\boldsymbol{x}_2 & \boldsymbol{x}_2^2 \\ \vdots & \vdots & \vdots \\ 1 & \sqrt{2}\boldsymbol{x}_n & \boldsymbol{x}_n^2 \end{bmatrix}$$

☹

*We must compute the feature maps **explicitly**.*

Not practical for high-dimensional or infinite-dimensional spaces.

What If We Could Parameterize $\hat{\boldsymbol{\theta}}$ and the Estimation y?

Given a set of data points $\{\boldsymbol{x}_i, y_i\}_{i=1}^n$, one may re-write the parameter as

$$\hat{\boldsymbol{\theta}} = \sum_{i=1}^{n} \hat{\alpha}_i \, \Phi(\boldsymbol{x}_i)$$

For some $\hat{\alpha} \in \mathbb{R}^n$ such that

$$y = \boldsymbol{\theta}^\top \Phi(\boldsymbol{x}) = \sum_{i=1}^{n} \hat{\alpha}_i \, \Phi(\boldsymbol{x}_i)^\top \Phi(\boldsymbol{x})$$

Hence, we won't need to compute the feature map $\Phi(\boldsymbol{x})$ explicitly. We only need to compute the **inner product** between $\Phi(\boldsymbol{x}_i)$ and $\Phi(\boldsymbol{x})$.

Kernel Regression

$$\Phi(X) = \begin{bmatrix} 1 & \sqrt{2}x_1 & x_1^2 \\ 1 & \sqrt{2}x_2 & x_2^2 \\ \vdots & \vdots & \vdots \\ 1 & \sqrt{2}x_n & x_n^2 \end{bmatrix}$$

$$\Phi(\boldsymbol{x}_i)^\top \Phi(\boldsymbol{x}) = 1 + 2\boldsymbol{x}\boldsymbol{x}_i + \boldsymbol{x}^2\boldsymbol{x}_i^2 \quad := k(\boldsymbol{x}, \boldsymbol{x}_i)$$

kernel trick!

$$\hat{y} = \sum_{i=1}^{n} \hat{\alpha}_i \, k(\boldsymbol{x}_i, \boldsymbol{x})$$

Kernel

Computes the inner product (a similarity measure) between two vectors in a feature space.

We can express inner products of nonlinear function of the original input space.

How to Parameterize $\hat{\boldsymbol{\theta}} = \sum_{i=1}^{n} \hat{\alpha}_i \, \Phi(\boldsymbol{x}_i)$?

Lemma:

$$(\boldsymbol{X}^\top \boldsymbol{X} + \lambda \boldsymbol{I})^{-1} \boldsymbol{X}^\top = \boldsymbol{X}^\top (\boldsymbol{X}\boldsymbol{X}^\top + \lambda \boldsymbol{I})^{-1}$$

We can re-write the estimation

$$\hat{\boldsymbol{\theta}} = (\Phi(\boldsymbol{X})^\top \Phi(\boldsymbol{X}) + \lambda \boldsymbol{I})^{-1} \Phi(\boldsymbol{X})^\top \boldsymbol{y}$$

as

$$\hat{\boldsymbol{\theta}} = \Phi(\boldsymbol{X})^\top \underbrace{(\Phi(\boldsymbol{X})\Phi(\boldsymbol{X})^\top + \lambda \boldsymbol{I})^{-1} \boldsymbol{y}}_{\hat{\boldsymbol{\alpha}}}$$

$$\hat{\boldsymbol{\theta}} = \begin{bmatrix} \Phi(\boldsymbol{x}_1) \\ \Phi(\boldsymbol{x}_2) \\ \vdots \\ \Phi(\boldsymbol{x}_n) \end{bmatrix}^\top \begin{bmatrix} \hat{\alpha}_1 \\ \hat{\alpha}_2 \\ \vdots \\ \hat{\alpha}_n \end{bmatrix} = \sum_{i=1}^{n} \hat{\alpha}_i \, \Phi(\boldsymbol{x}_i)$$

Kernel Regression Remarks

(I) Given the kernel $k(\,\cdot\,,\cdot\,)$, for prediction for a new point $\boldsymbol{x}$, we have

$$\hat{y} = \boldsymbol{\theta}^\top \Phi(\boldsymbol{x}) = \sum_{i=1}^{n} \hat{\alpha}_i \; \Phi(\boldsymbol{x}_i)^\top \Phi(\boldsymbol{x}) = \sum_{i=1}^{n} \hat{\alpha}_i \; k(\boldsymbol{x}_i, \boldsymbol{x})$$

(II) The kernel weights $\hat{\alpha}_i$ is given by

$$\hat{\alpha} = (\Phi(\boldsymbol{X})\Phi(\boldsymbol{X})^\top + \lambda \boldsymbol{I})^{-1}\boldsymbol{y} = (\boldsymbol{K} + \lambda \boldsymbol{I})^{-1}\boldsymbol{y}$$

$$\boldsymbol{K} \in \mathbb{R}^{n\times n},\; k_{ij} = \Phi(\boldsymbol{x}_i)^\top \Phi(\boldsymbol{x}_j) = k(\boldsymbol{x}_i, \boldsymbol{x}_j)$$

An Alternate Formulation for Kernel Regression

Given a known kernel $k(\,\cdot\,,\cdot\,)$, what is the unknown parameter below?

$$\hat{y} = \sum_{i=1}^{n} \hat{\alpha}_i \; k(\boldsymbol{x}_i, \boldsymbol{x})$$

$$\hat{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}} \; \frac{1}{2}\|\boldsymbol{y} - \Phi(\boldsymbol{X})\boldsymbol{\theta}\|_2^2 \; + \; \frac{\lambda}{2}\|\boldsymbol{\theta}\|_2^2$$

It's $\hat{\boldsymbol{\alpha}}$! so, we can optimization for $\boldsymbol{\theta}$ as a function of $\boldsymbol{\alpha}$

$$\hat{\boldsymbol{\alpha}} = \arg\min_{\boldsymbol{\alpha}} \; \frac{1}{2}\|\boldsymbol{y} - \Phi(\boldsymbol{X})(\textstyle\sum_{i=1}^{n} \alpha_i \; \Phi(\boldsymbol{x}_i))\|_2^2 \; + \; \frac{\lambda}{2}\|\textstyle\sum_{i=1}^{n} \alpha_i \; \Phi(\boldsymbol{x}_i)\|_2^2$$

$$\hat{\boldsymbol{\alpha}} = \arg\min_{\boldsymbol{\alpha}} \; \frac{1}{2}\|\boldsymbol{y} - \textstyle\sum_{i=1}^{n} \alpha_i \; k(\boldsymbol{x}, \boldsymbol{x}_i)\|_2^2 \; + \; \frac{\lambda}{2}\boldsymbol{\alpha}^\top \boldsymbol{K}\boldsymbol{\alpha}$$

Properties of Kernels

Symmetric

$$\langle \Phi(\boldsymbol{x}), \Phi(\boldsymbol{y}) \rangle = \langle k(\,\cdot\,, \boldsymbol{x}), k(\,\cdot\,, \boldsymbol{y}) \rangle = k(\boldsymbol{x}, \boldsymbol{y}) = k(\boldsymbol{y}, \boldsymbol{x})$$

Positive Definite (PD)

Let $k : S \times S \to \mathbb{R}$. The Kernel $k(\,\cdot\,,\cdot\,)$ is a positive definite kernel if

$$\sum_{i=1}^{n} \sum_{j=1}^{n} \alpha_i \alpha_j \; k(\boldsymbol{x}_i, \boldsymbol{y}_j) \geq 0 \quad \forall \boldsymbol{\alpha} \in \mathbb{R}^n, \quad \boldsymbol{x}_i, \boldsymbol{x}_j \in S$$

$$\boldsymbol{\alpha}^\top \boldsymbol{K} \boldsymbol{\alpha} \geq 0$$

Constructing Kernels

Let $k_1 : S \times S \to \mathbb{R}$ and $k_2 : S \times S \to \mathbb{R}$ be symmetric and positive definite kernels. Then, we can construct kernels that are symmetric and PD as follows:

$$k(\,\cdot\,,\cdot\,) = k_1(\,\cdot\,,\cdot\,) + k_2(\,\cdot\,,\cdot\,)$$

$$k(\,\cdot\,,\cdot\,) = c\; k_1(\,\cdot\,,\cdot\,), \quad c > 0$$

$$k(\,\cdot\,,\cdot\,) = k_1(\,\cdot\,,\cdot\,)\; k_2(\,\cdot\,,\cdot\,)$$

Example Kernels

Kernels quantify some notion of similarity (inner product of their feature maps) between two vectors (data examples).

Linear Kernel

$$k(\boldsymbol{x}, \boldsymbol{y}) = \boldsymbol{x}^\top \boldsymbol{y}$$

You don't need to explicitly construct that feature space.

Homogeneous Polynomial

$$k(\boldsymbol{x}, \boldsymbol{y}) = (\boldsymbol{x}^\top \boldsymbol{y})^d, \; d \in \mathbb{N}^+$$

Inhomogeneous Polynomial

$$k(\boldsymbol{x}, \boldsymbol{y}) = (c + \boldsymbol{x}^\top \boldsymbol{y})^d, \; d \in \mathbb{N}^+$$

Gaussian Kernel

$$k(\boldsymbol{x}, \boldsymbol{y}) = e^{\frac{-\|\boldsymbol{x} - \boldsymbol{y}\|_2^2}{2\sigma^2}}$$

Feature Map Corresponding to Inhomogeneous Kernel Example

Let $k : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ be defined as $k(x, y) = (1 + xy)^2$. Then, find its corresponding feature map.

$$k(x, y) \;=\; (1 + xy)^2 = 1 + 2xy + x^2y^2 \;=\; [1 \;\sqrt{2}x \;\; x^2] \begin{bmatrix} 1 \\ \sqrt{2}y \\ y^2 \end{bmatrix}$$

$$\Phi(x) = \begin{bmatrix} 1 \\ \sqrt{2}x \\ x^2 \end{bmatrix}$$

What is Happening in Regression to Fit and Predict the Data?

Consider the **linear kernel** $k : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$, defined as $k(x, y) = xy$.

$$\langle \Phi(\boldsymbol{x}), \Phi(\boldsymbol{y}) \rangle = \langle k(\,\cdot\,, \boldsymbol{x}), k(\,\cdot\,, \boldsymbol{y}) \rangle = k(\boldsymbol{x}, \boldsymbol{y}) = k(\boldsymbol{y}, \boldsymbol{x})$$

$$\hat{y} = \sum_{i=1}^{n} \hat{\alpha}_i \, k(\boldsymbol{x}_i, \boldsymbol{x})$$

Intuition: closer the feature $\boldsymbol{x}$ to $\boldsymbol{x}_i$, then $k(\boldsymbol{x}_i, \boldsymbol{x})$ is larger, hence the parameter α_i will have higher impact in prediction of y.

Gaussian Kernels for Regression

Given a set of data points $\{\boldsymbol{x}_i, y_i\}_{i=1}^n$, and the Gaussian kernel, we regress as

$$\hat{y} = \sum_{i=1}^n \alpha_i \; e^{\frac{-1}{2\sigma^2}\|\boldsymbol{x}-\boldsymbol{x}_i\|_2^2}$$

Intuitively, what is happening?

Let get on the board.

This is also known as
radial basis function (**RBF**).

Kernels Approach for Interpretability Analysis

Goal: represent data $\boldsymbol{x}$ as linear combination of set of features

$$\boldsymbol{x}^i = \boldsymbol{D}\boldsymbol{z}^i$$

$$\min_{\boldsymbol{D},\boldsymbol{z}} \sum_{i=1}^{n} \|\boldsymbol{x}^i - \boldsymbol{D}\boldsymbol{z}^i\|_2^2$$

The optimal dictionary can be represented as linear combination of data examples, i.e.,

$$\boldsymbol{D}_j = \sum_{i=1}^{n} \tau_i \boldsymbol{x}^i \quad \Rightarrow \quad \hat{\boldsymbol{x}}_{\text{new data}} = \sum_{i=1}^{n} \beta_i \boldsymbol{x}^i$$

Published in Transactions on Machine Learning Research (08/2022)

Stable and Interpretable Unrolled Dictionary Learning

Bahareh Tolooshams btolooshams@seas.harvard.edu
Demba Ba demba@seas.harvard.edu
School of Engineering and Applied Sciences
Harvard University

Reviewed on OpenReview: https://openreview.net/forum?id=e3S0Bl2R08

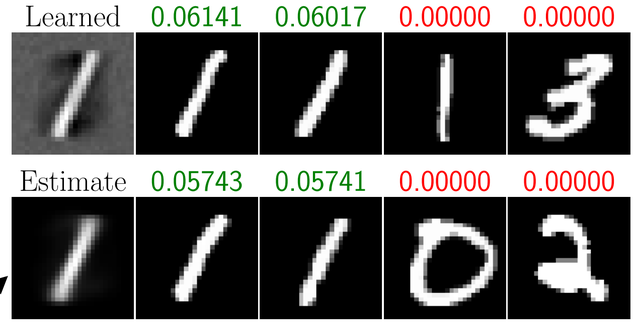
How is your model performing a particular prediction?

Example for Kernel Perspective for Interpretability

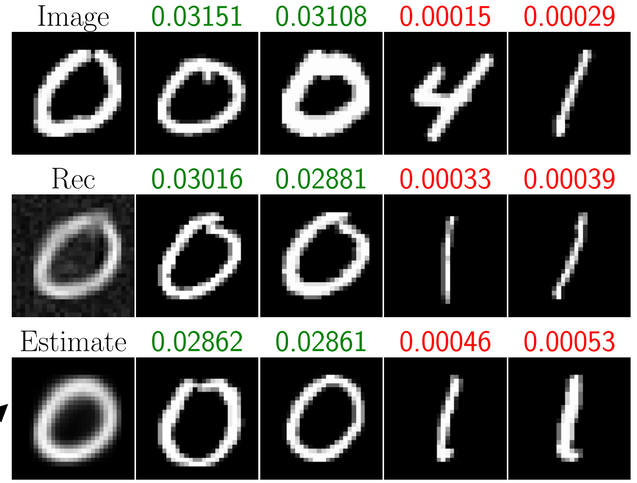
MNIST dataset

Learned dictionary model

$$\boldsymbol{D}_j^{\text{learned}} = \sum_{i=1}^{n} \tau_i \boldsymbol{x}^i$$



$$\hat{\boldsymbol{x}}^{\text{test}} = \sum_{i=1}^{n} \beta_i \boldsymbol{x}^i$$



Why is a particular feature learned?

How is an image reconstructed based on training images?

Takeaways

- *With **kernels**, no need for **explicit** construction of the feature matrix.*

- *With **kernels**, you can think of your regression as linear weighted sum of feature similarities.*